Project Instructions for Students: -

The graduation project is a key requirement for obtaining the Digital Egypt Pioneers Initiative Completion Certificate.

- Students are free to choose any of the ideas listed in the project booklet for their respective career track without any restrictions **"With the management of the initiative being duly informed."**, they are able to choose other ideas not listed in the booklet, but it should go in the same format of the ideas given.
- The project is a group assignment, and teams should consist of 4 to 6 students.
- Within a maximum of one week from the announcement of the project booklet, students must form their groups and inform the instructor. If they fail to do so, the instructor has the right to assign groups randomly and announce the team members.
- Students must divide the work responsibilities within the group and inform the instructor within two weeks of the project booklet announcement. During the final presentation, each group must demonstrate the work completed and each member's responsibility for their assigned tasks.
- The final evaluation will be based on the final presentation, which must include the students' adherence to the deliverables and the distribution of tasks among team members.

تعليمات المشروع للطلاب:- مشروع التخرج هو أحد المتطلبات الأساسية للحصول على شهادة

إتمام مبادرة رواد مصر الرقمية.

- يتمتع الطلاب بحرية اختيار أي من الأفكار المدرجة في كتيب المشروع لمسارهم الوظيفي دون أي قيود، أو اختيار أي فكره أخرى غير مدرجه **)مع اعلام إدارة المبادرة بها(**، ولكن بنفس الطريقة المستخدمة في الأفكار المذكورة .
- المشروع عمل جماعي، ويجب أن تتكون فرق العمل من 4 إلى 6 طلاب.
- في غضون أسبوع كحد أقصى من إعلان كتيب المشروع، يجب على الطلاب تشكيل فرقهم وإبلاغ المدرب بذلك. في حالة عدم القيام بذلك، يحق للمدرب تقسيمهم بشكل عشوائي وإعلان أعضاء الفريق.
- يجب على الطلاب تقسيم مسؤوليات العمل داخل المجموعة وإبلاغ المدرب بها في غضون أسبوعين من إعلان كتيب المشروع. كما يجب على كل مجموعة خلال العرض النهائي توضيح الأعمال التي تم إنجازها وتحديد مسؤولية كل فرد في تنفيذها.
- سيتم التقييم النهائي بناءً على العرض النهائي، والذي يجب أن يتضمن التزام الطلاب بتسليم المخرجات وتقسيم العمل بين أعضاء الفريق.

Project 2: Sales Forecasting and Optimization

Project Overview:

The **Sales Forecasting and Optimization** project aims to predict future sales for a retail or e-commerce business by using historical sales data. The project involves data collection, cleaning, exploration, time-series forecasting model development, optimization, and deployment. The end goal is to have a model that can generate accurate sales predictions to help businesses optimize inventory, marketing, and sales strategies.

Milestone 1: Data Collection, Exploration, and Preprocessing

Objectives:

- Collect and explore historical sales data and preprocess it for analysis and model building.

Tasks:

1. **Data Collection:**
   - Acquire a dataset containing historical sales data (e.g., daily or weekly sales data from retail or e-commerce platforms).
   - Ensure that the dataset includes relevant features like sales amount, date, promotions, holidays, weather, etc.
2. **Data Exploration:**
   - Perform exploratory data analysis (EDA) to understand trends, seasonality, and missing values in the dataset.
   - Generate summary statistics, check for outliers, and identify key patterns and correlations (e.g., sales with promotions, holidays).
3. **Data Preprocessing:**
   - Handle missing values, remove duplicates, and address any data inconsistencies.
   - Engineer time-based features (e.g., day of the week, month, seasonality, promotional periods).
   - Apply data scaling and transformations (e.g., normalization) as needed for modeling.

Deliverables:

- **Data Exploration Report:** A summary of findings from data exploration, including trends, seasonality, and any data quality issues.
- **Exploratory Data Analysis (EDA) Notebook:** A Jupyter notebook with visualizations such as line plots, histograms, and correlation heatmaps to reveal key insights.

- **Cleaned Dataset:** A well-processed dataset that is ready for analysis and modeling.

Milestone 2: Data Analysis and Visualization

Objectives:

- Clean and preprocess the data further and visualize the relationships in the data.

Tasks:

1. **Data Cleaning:**
   - Address any remaining missing values, outliers, and inconsistencies in the dataset.
2. **Data Analysis:**
   - Perform statistical analysis to identify correlations between sales and other factors such as promotions, weather, holidays, and special events.
   - Investigate seasonality and long-term trends in sales.
3. **Data Visualization:**
   - Create visualizations like line graphs, bar charts, and scatter plots to display sales trends and seasonal patterns.
   - Develop interactive dashboards (using Plotly or Dash) to allow users to explore trends and patterns in the sales data.

Deliverables:

- **Cleaned Dataset and Analysis Report:** A report documenting the data cleaning steps, challenges, and insights from the analysis.
- **Advanced Visualizations:** Interactive visualizations (e.g., time series trends, seasonal patterns) or dashboards to explore the data.

Milestone 3: Forecasting Model Development and Optimization

Objectives:

- Build and optimize forecasting models to predict future sales.

Tasks:

1. **Model Selection:**
   - Choose appropriate time-series forecasting models (e.g., ARIMA, SARIMA, Facebook Prophet, XGBoost, or LSTM if applicable).
2. **Model Training:**
   - Split the dataset into training and test sets, ensuring proper time-series validation techniques (e.g., rolling-window, train-test split).

- Train multiple models and assess their performance using error metrics like RMSE (Root Mean Squared Error), MAE (Mean Absolute Error), and MAPE (Mean Absolute Percentage Error).

3. **Model Evaluation and Tuning:**
   - Tune hyperparameters for selected models (e.g., using Grid Search, Random Search, or Bayesian Optimization).
   - Evaluate residuals to ensure that no patterns are left unmodeled.
4. **Model Selection:**
   - Compare the models' performance and choose the best-performing model based on evaluation metrics.

Deliverables:

- **Forecasting Model Performance Report:** A report summarizing model performance, key metrics (RMSE, MAE, etc.), and the chosen model.
- **Model Code:** Python code used for training, evaluating, and optimizing forecasting models.
- **Final Forecasting Model:** The final selected model with optimized hyperparameters ready for deployment.

Milestone 4: MLOps, Deployment, and Monitoring

Objectives:

- Implement MLOps for model tracking and deploy the forecasting model for real-time or batch predictions.

Tasks:

1. **MLOps Implementation:**
   - Use tools like **MLflow** to track experiments, manage models, and log metrics and parameters.
   - Implement version control for models and datasets using tools like **DVC** (Data Version Control).
2. **Deployment:**
   - Deploy the model using frameworks such as **Flask** or **Streamlit** to provide a user interface for generating real-time sales forecasts.
   - Ensure that the model can handle batch or real-time predictions depending on business needs.
   - Optionally, deploy to a cloud platform (e.g., Google Cloud, AWS, or Heroku) for scalable deployment.
3. **Model Monitoring:**

- Set up model monitoring to track performance over time and detect issues like model drift.
- Establish a feedback loop for continuous model improvement based on prediction accuracy.

4. **Performance Reporting:**
    - Log model performance and set up alert systems to notify stakeholders if prediction accuracy drops below a defined threshold.

Deliverables:

- **Deployed Model:** A live sales forecasting model deployed to a web app or cloud platform for realtime or batch predictions.
- **MLOps Report:** A report documenting the tools and processes used to manage the forecasting model, including experiment tracking, model versioning, and deployment pipeline.
- **Monitoring Setup:** A detailed setup explaining how the model's performance is being tracked and maintained over time.

Milestone 5: Final Documentation and Presentation

Objectives:

- Document the entire process and prepare a presentation for stakeholders to highlight the impact and business value of the project.

Tasks:

1. **Final Report:**
    - Summarize the entire project, including data exploration, model development, optimization, and deployment.
    - Discuss insights derived from the analysis, the business implications of accurate sales forecasting, and how it can optimize sales and inventory strategies.
    - Highlight challenges faced during the project and how they were overcome.
2. **Final Presentation:**
    - Prepare a concise and engaging presentation to explain the methodology, results, and business value of the forecasting model.
    - Demonstrate the deployed model’s functionality and showcase its ability to generate realtime or batch sales forecasts.
    - Discuss potential use cases for the model in optimizing sales strategies and suggest areas for future improvements.

Deliverables:

- **Final Project Report:** A comprehensive document summarizing all project steps, from data collection to deployment, with insights into the model's impact on business operations.

- **Final Presentation:** A well-structured presentation (e.g., PowerPoint or Google Slides) for stakeholders, demonstrating the forecasting model's capabilities and value.

Final Milestones Summary:

Milestone	Key Deliverables
1. **Data Collection, Exploration & Preprocessing**	EDA Report, Interactive Visualizations, Cleaned Dataset
2. **Data Analysis, Visualization & Feature Engineering**	Data Analysis Report, Enhanced Visualizations, Feature Engineering Summary
3. **Model Development & Optimization**	Model Evaluation Report, Model Code, Final Model
4. **MLOps, Deployment & Monitoring**	Deployed Model, MLOps Report, Monitoring Setup
5. **Final Documentation & Presentation**	Final Project Report, Final Presentation

---

Conclusion:

The **Sales Forecasting and Optimization** project leverages historical sales data to build a robust forecasting model that helps businesses predict future sales trends. This structured approach ensures that the model is not only accurate but also deployable and sustainable in a production environment, making it an invaluable tool for sales optimization.

Tasks:

1. **Final Report:**
   - Summarize the entire project, including data collection, preprocessing, model development, and deployment.
   - Discuss challenges faced during the project and key insights gained from the predictive model.
   - Provide recommendations for how healthcare professionals can integrate the model into their workflow to improve patient outcomes.
2. **Final Presentation:**
   - Create a concise and engaging presentation for healthcare stakeholders, showcasing the predictive model's functionality and real-world impact.
   - Discuss the model’s ability to predict patient risk, identify trends in health data, and assist healthcare professionals in decision-making.
   - Highlight potential future improvements and how the model can evolve with more data and integration into healthcare systems.

Deliverables:

- **Final Project Report:** A comprehensive document summarizing all aspects of the project, including the model's impact on healthcare outcomes.
- **Final Presentation:** A visually engaging presentation suitable for healthcare stakeholders, demonstrating the model's functionality and business implications.

Final Milestones Summary:

Milestone	Key Deliverables
1. **Data Collection, Exploration & Preprocessing**	EDA Report, Interactive Visualizations, Cleaned Dataset
2. **Data Analysis, Visualization & Feature Engineering**	Data Analysis Report, Visualizations of Health Trends, Feature Engineering Summary
3. **Model Development & Optimization**	Model Evaluation Report, Model Code, Final Model
4. **MLOps, Deployment & Monitoring**	Deployed Model, MLOps Report, Monitoring Setup
5. **Final Documentation & Presentation**	Final Project Report, Final Presentation

---

Conclusion:

Tasks:

1. **Final Report:**
   - Provide a comprehensive summary of the project, including the problem definition, data exploration, feature engineering, and model development.
   - Discuss how the forecasting model can optimize sales and inventory management, improve demand planning, and inform marketing and staffing decisions.
   - Highlight challenges faced, decisions made, and the model's business impact.
2. **Final Presentation:**
   - Create a presentation that demonstrates the value of the forecasting model for sales and demand prediction.
   - Include a live demo or walkthrough of the deployed model showing how business stakeholders can use it for decision-making.
3. **Future Improvements:**
   - Suggest potential improvements, such as incorporating more external features (e.g., competitor activity, macroeconomic data), testing alternative algorithms (e.g., Prophet), or enhancing deployment capabilities.

Deliverables:

- **Final Project Report**: A detailed summary of the project, from data collection to deployment, and its business impact.
- **Final Presentation**: A polished presentation for business stakeholders, explaining the forecasting model's value and usage.

Final Milestones Summary:

Milestone	Key Deliverables
1. Data Collection, Exploration & Preprocessing	EDA Report, Interactive Visualizations, Cleaned Dataset
2. Advanced Data Analysis, Visualization & Feature Engineering	Data Analysis Report, Enhanced Visualizations, Feature Engineering Summary
3. Model Development & Optimization	Model Evaluation Report, Model Code, Final Model
4. MLOps, Deployment & Monitoring	Deployed Model, MLOps Report, Monitoring Setup
5. Final Documentation & Presentation	Final Project Report, Final Presentation

---

Conclusion:

The **Sales Forecasting and Optimization** project leverages historical sales data to build a robust forecasting model that helps businesses predict future sales trends. This structured approach ensures that the model is not only accurate but also deployable and sustainable in a production environment, making it an invaluable tool for sales optimization.